EC 571: Digital VLSI Circuit Design

MOS Transistor Model

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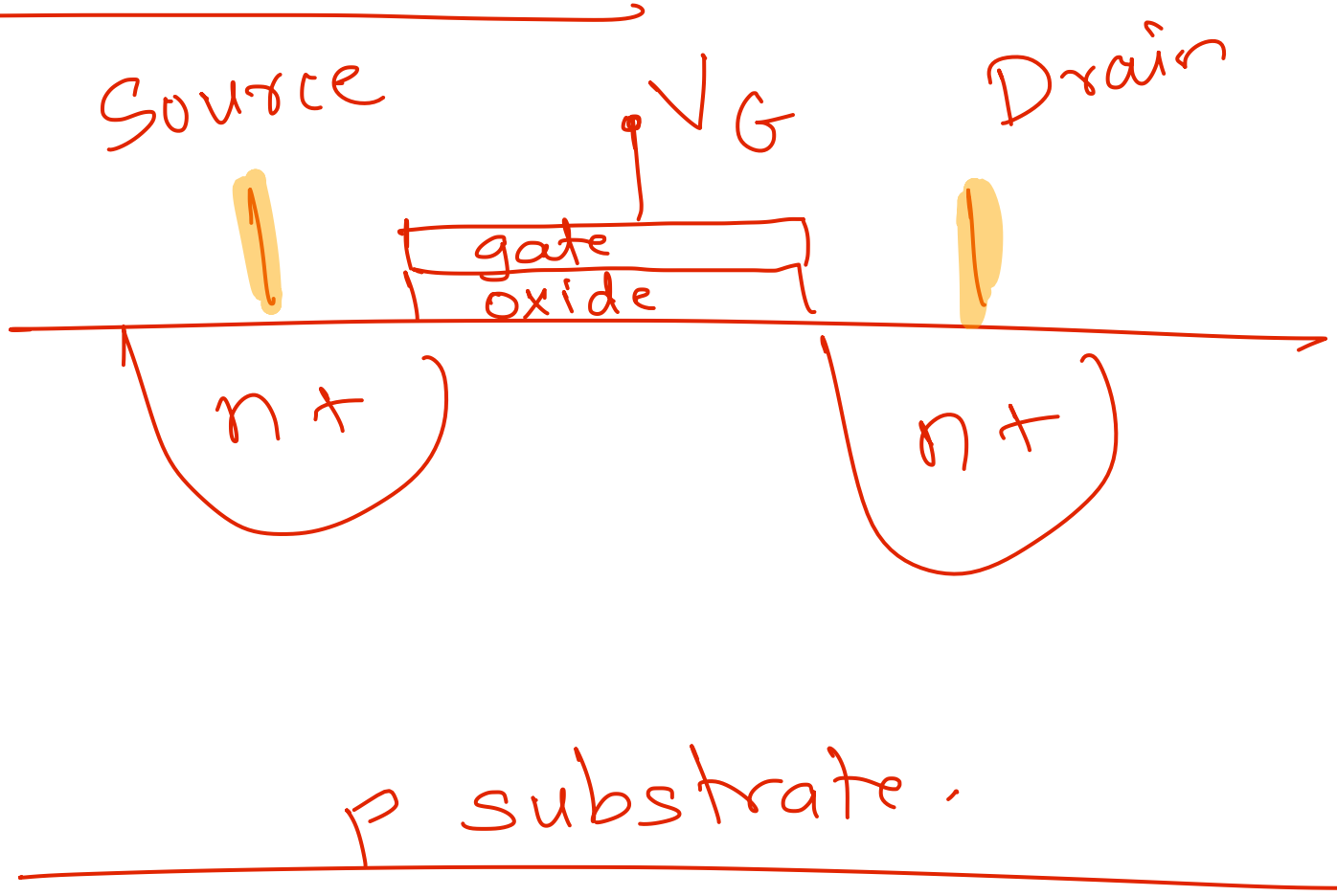
Acknowledgement - Some slides adapted from:
- J. Rabaey et al. Digital Integrated Circuits 2nd Ed.
- UC Berkeley EE141, UC Berkeley EE241, UC Berkeley EE251
- Stanford EE271

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Last lecture

- Last lecture
  - Gate delay and Logical effort
- Today's lecture
  - MOS Transistor model
- Reading
  - Chapter 3

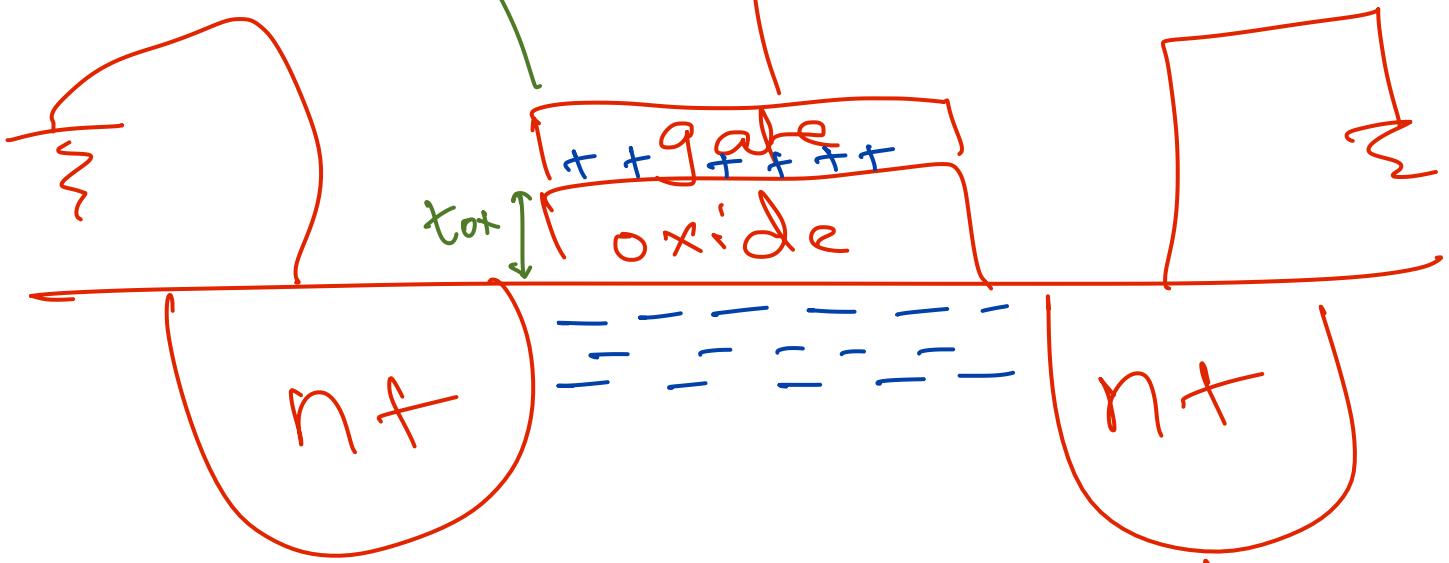
MOS transistor



Case I Source, Drain & Bulk/Body are grounded & +ve voltage applied to Gate

$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$

V_G +ve voltage



p-substrate.

Body

$V_G = 2\phi_F \simeq 0.6$

↑ Fermi potential.

Critical voltage at which we have a channel between source and drain is the threshold voltage

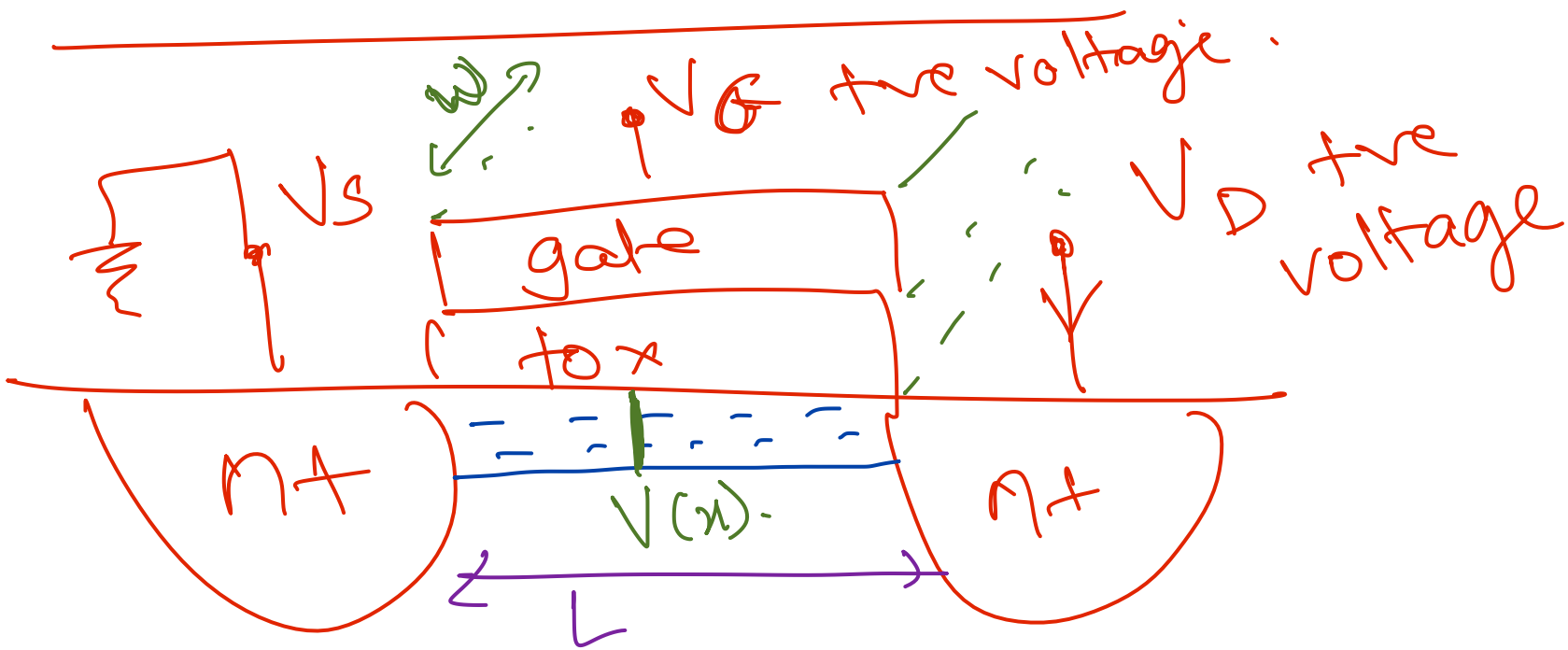
$V_T \Rightarrow$ Threshold voltage.

$$V_T = V_{T0} + \gamma\left(\sqrt{|-2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|}\right)$$

γ — body effect co-efficient

Not the same γ we had in the logical effort topic.

Case II Apply drain bias



$$I_D = Q(x)\, \mu_n\, W$$

- $Q(x)$: charge density.
- μ_n: drift velocity of the carriers
- W: width of the xtor.

$$Q = CV$$

$$= C_{ox}\left(\underbrace{V_{GS} - V(x)}_{\text{Voltage across the gate \& the channel.}} - V_{TH}\right)$$

$$v_n(x) = \mu E$$

mobility, electric field.

$$= \mu \frac{dV}{dx}.$$

$$I_D = W \cdot C_{ox} (V_{GS} - V(x) - V_{TH}) \cdot \mu \frac{dV}{dx}$$

$$I_D \, dx = W \cdot C_{ox} (V_{GS} - V(x) - V_{TH}) \mu \, dV$$

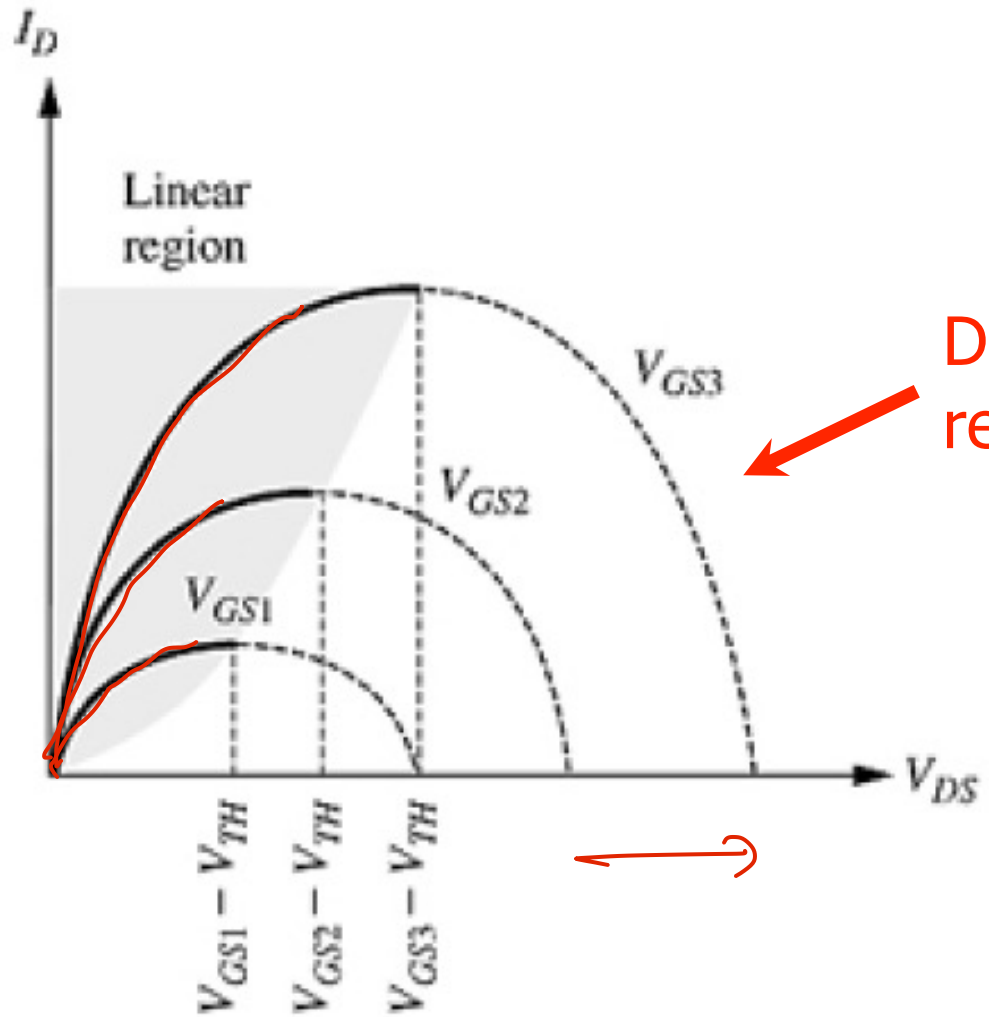
$$\int_0^L I_D \, dx = \int_0^{V_{DS}} W \cdot C_{ox} (V_{GS} - V(x) - V_{TH}) \mu \, dV$$

$$I_D = \frac{W}{L} \mu C_{ox} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{V_{DS}}{2} \right]$$

$\mu \cdot C_{ox} = k'_n$ = process transconductance.

$k'_n \cdot \frac{W}{L}$ = gain factor.

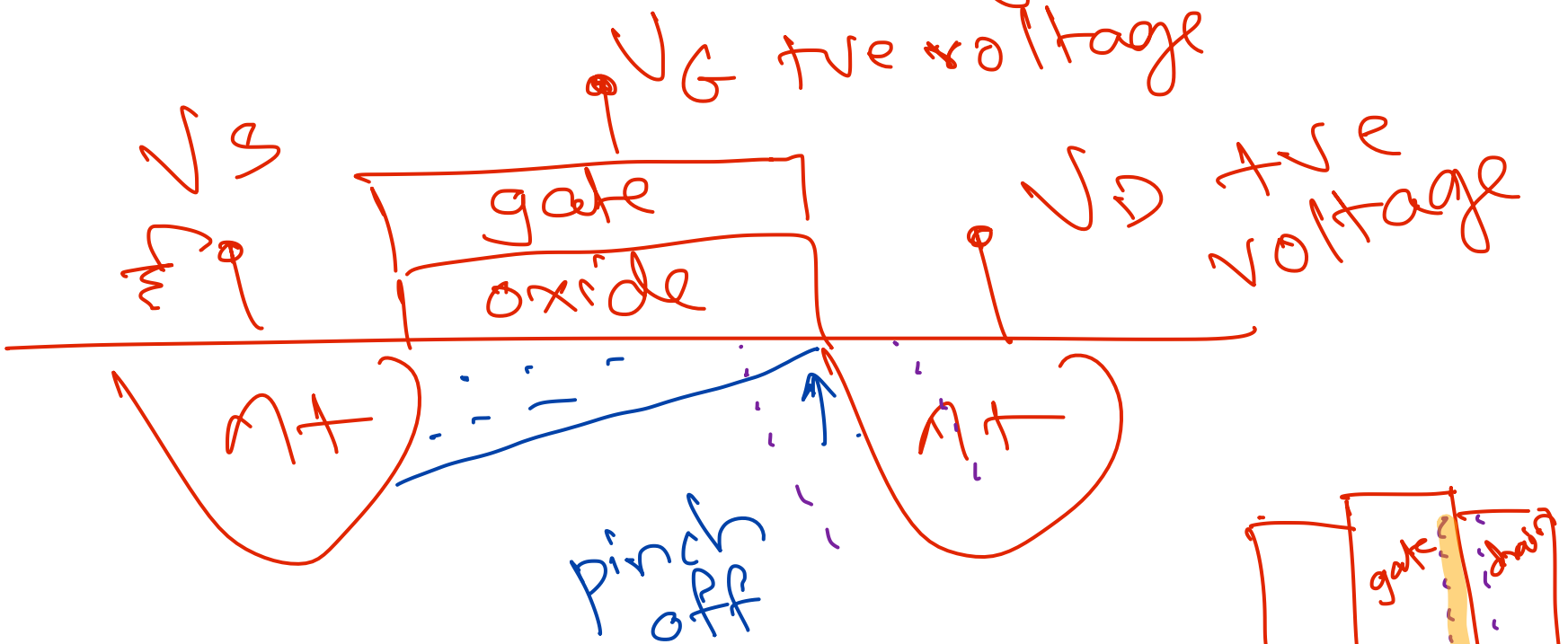
I_D versus V_{DS}

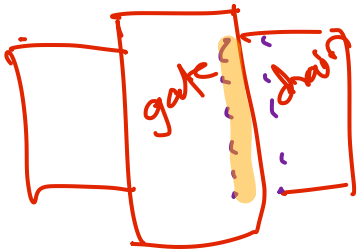


Does this really happen?

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As V_{DS} ↑, channel structure changes.





$V_{DS\,eff} = V_{GS} - V_{TH}$

xtor is in saturation

$$I_D = \frac{k'_n}{2} \frac{W}{L} (V_{GS} - V_{TH})^2$$

More precisely

$$I_D = \frac{k'_n}{2} \frac{W}{L} (V_{GS} - V_{TH})^2 \cdot (1 + \lambda V_{DS})$$

λ = channel length modulation parameter

Modes of operation

Cutoff : $V_{GS} - V_{TH} < 0 \quad I_D = 0$

Linear (Resistive) :

$$V_{GS} - V_{TH} \geq 0$$

$$V_{GS} - V_{TH} > V_{DS}$$

$$I_D = k'_n \frac{W}{L} \left[(V_{GS} - V_{TH}) \cdot V_{DS} - \frac{V_{DS}^2}{2} \right]$$

If $V_{DS} << V_{GS} - V_{TH}$

$$I_D = k'_n \frac{W}{L} \left[(V_{GS} - V_{TH}) \cdot V_{DS} \right]$$

$$\frac{V_{DS}}{I_D} = \frac{1}{k'_n \frac{W}{L} (V_{GS} - V_{TH})}$$

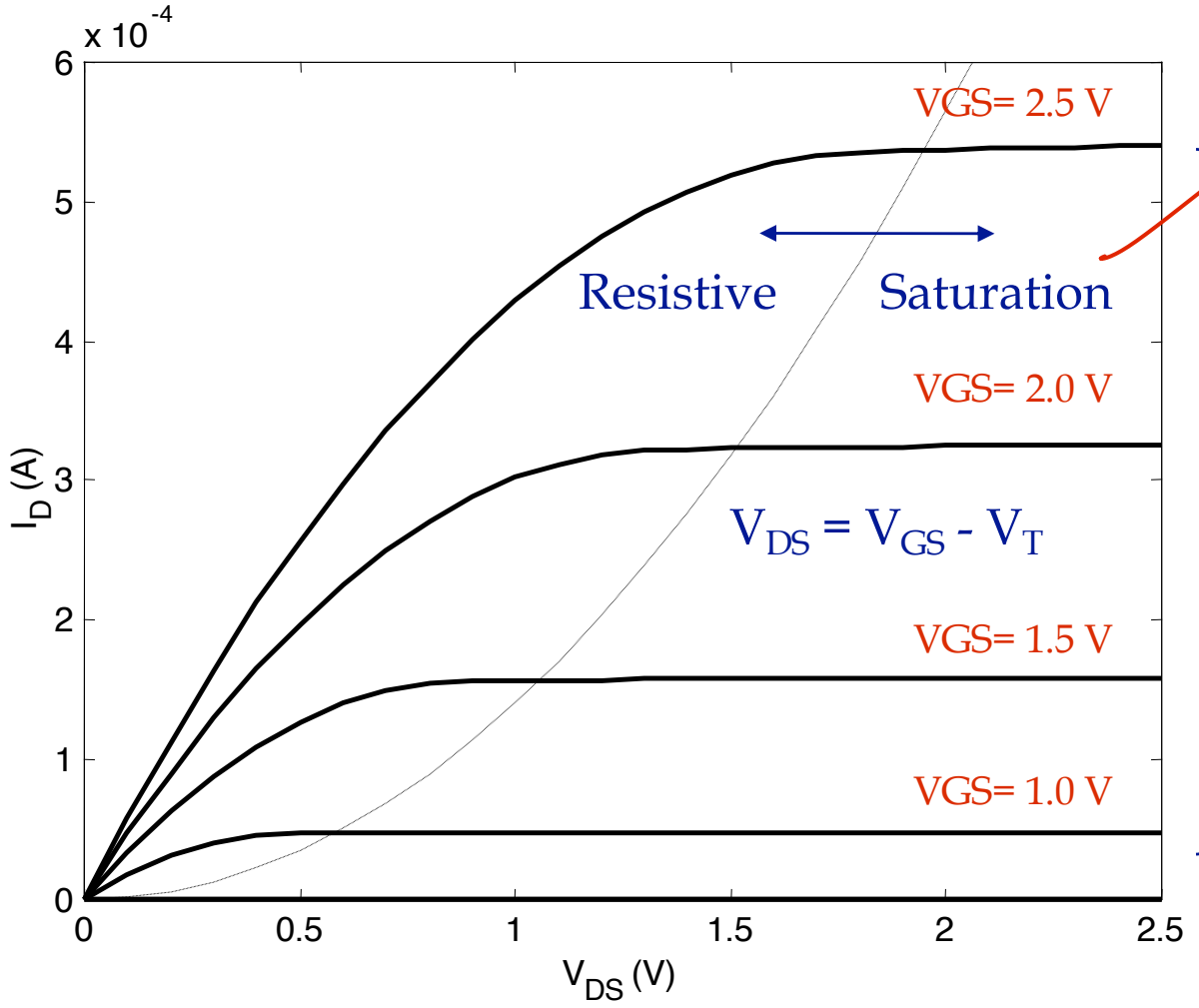
$$R_{eff} = \frac{1}{k'_n \frac{W}{L} (V_{GS} - V_{TH})} .$$

Saturation:

$$0 < V_{GS} - V_{TH} < V_{DS}$$

$$I_D = \frac{k'_n}{2} \frac{W}{L} (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS})$$

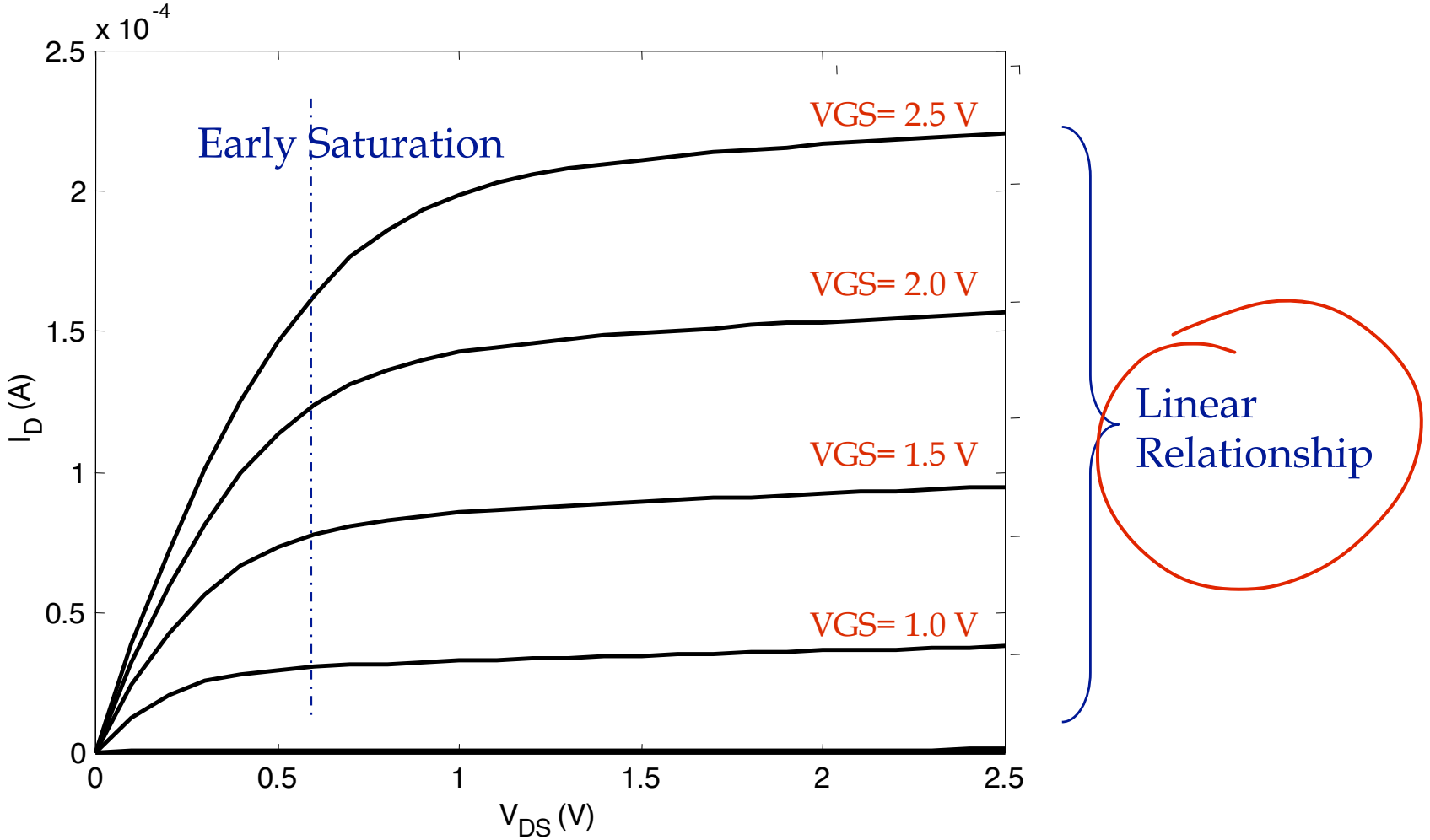
Current-Voltage Relations A good ol' transistor



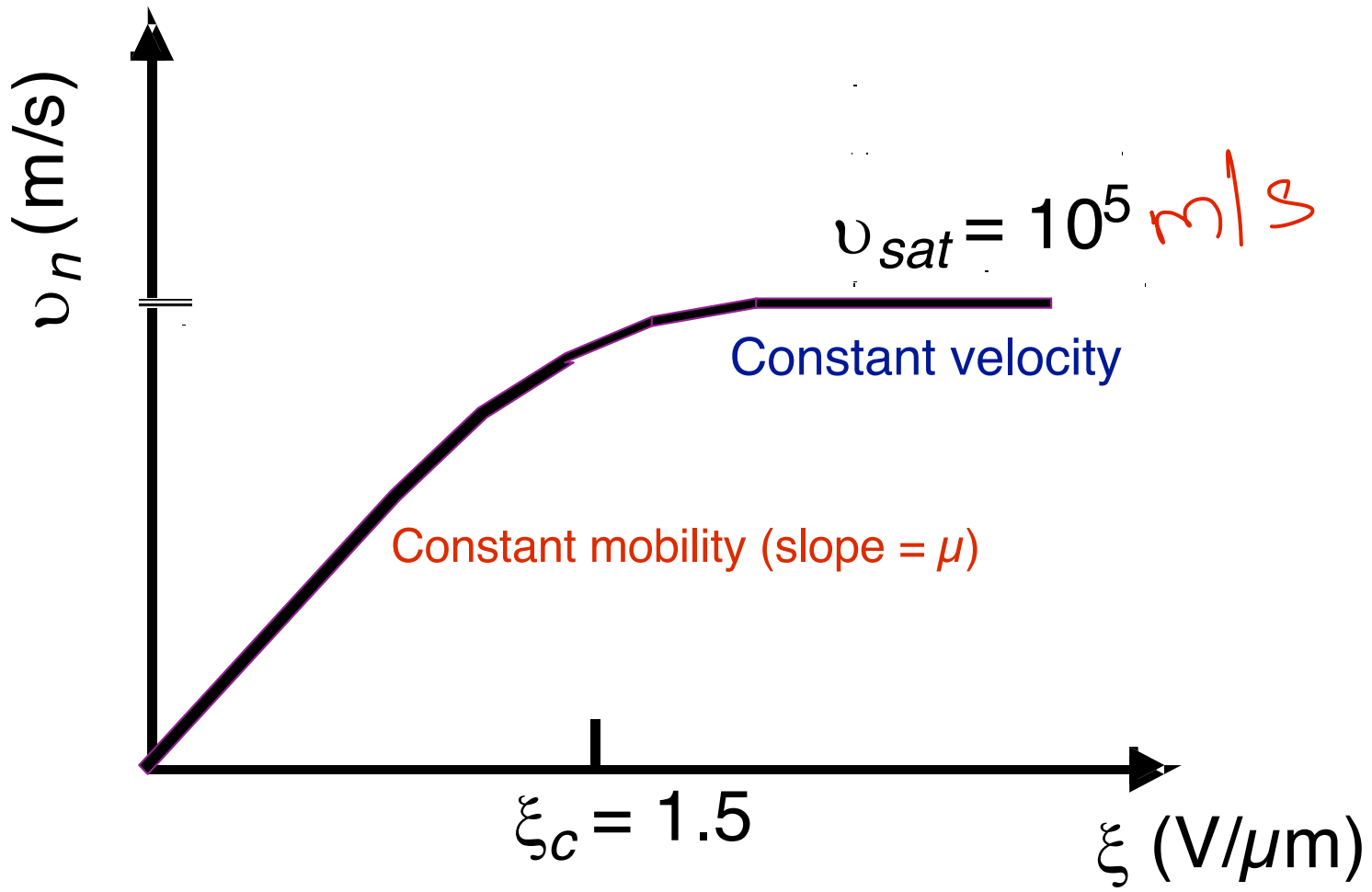
$I_D \propto (V_{GS} - V_{TH})^2$

Quadratic Relationship

Current-Voltage Relations - The Deep-Submicron Era

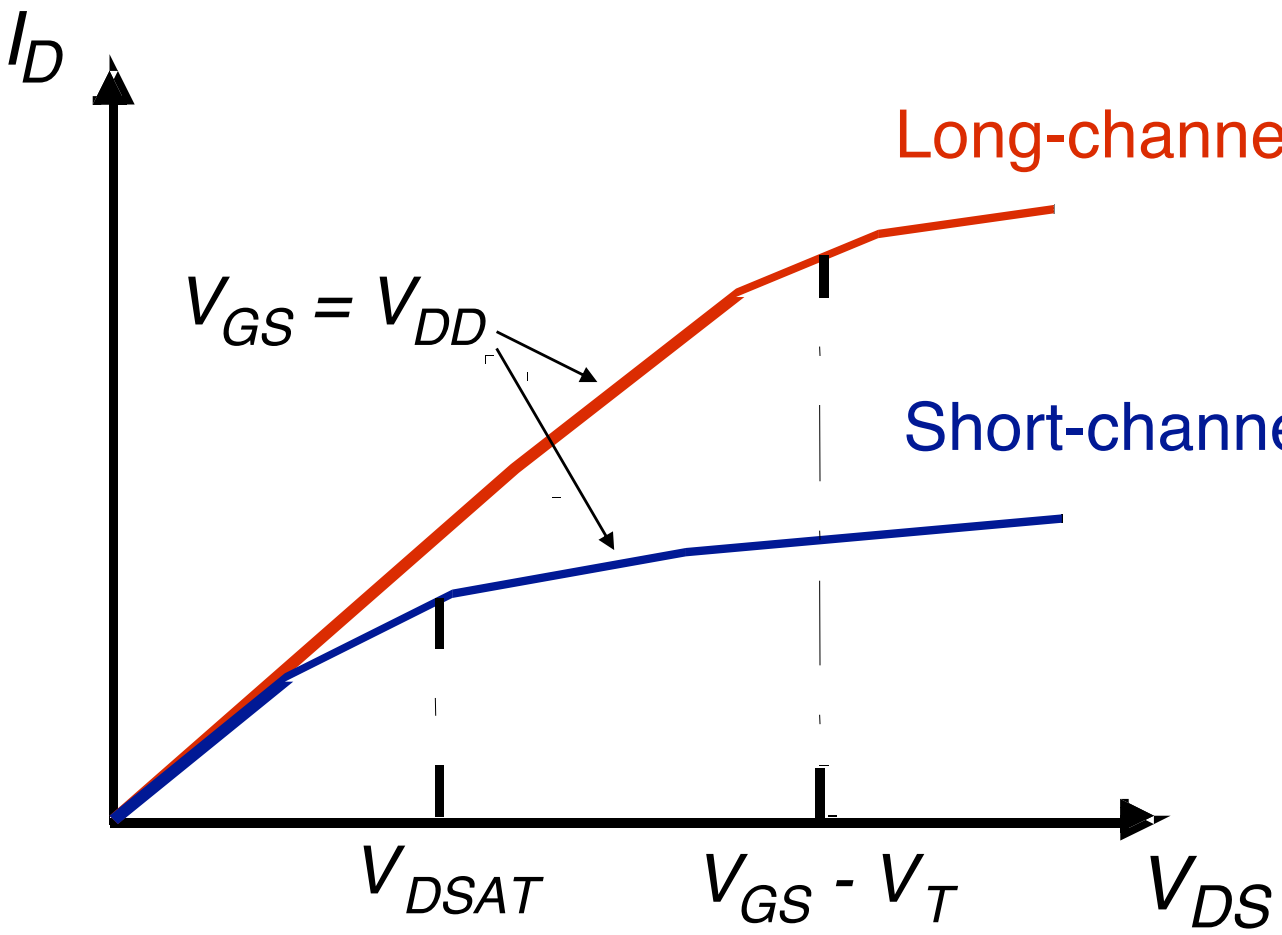


Velocity Saturation



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Perspective



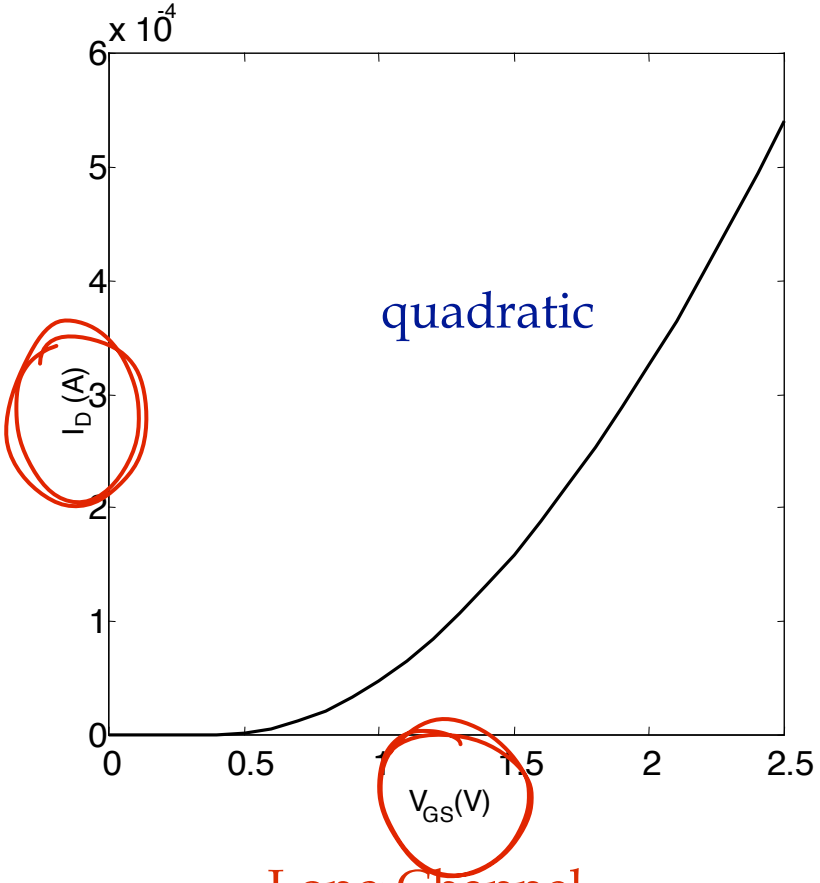
quadratic relationship

linear relationship

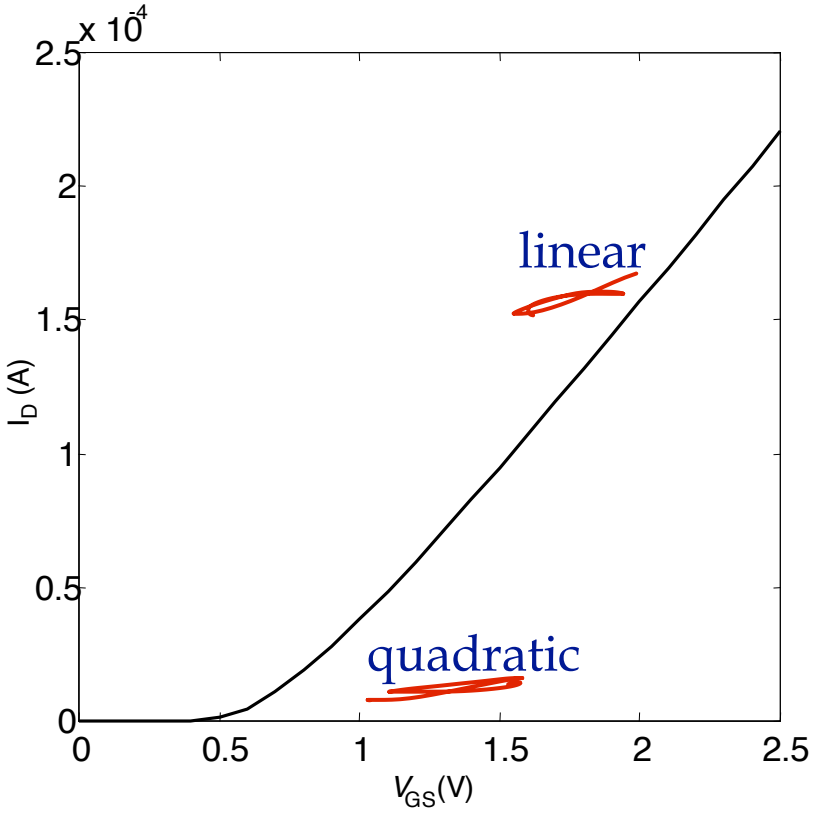
Velocity saturation.

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I_D versus V_{GS}



Long Channel



Short Channel

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Drain current in velocity saturation mode.

$$v = \frac{\mu \mathcal{E}}{1 + \mathcal{E}/\mathcal{E}_c} \quad \text{--- (1)}$$

← $\mathcal{E}_c$: critical electric field.

If $\mathcal{E} = \mathcal{E}_c$ --- (2)

$$v = v_{sat} \quad \text{--- (3)}$$

$$\mathcal{E}_c = \frac{2 v_{sat}}{\mu}$$

← Using (1), (2) & (3)

$$I_D = Q(x) \cdot v(x) \cdot W$$

$$I_D = \frac{\mu_n C_{ox}}{1 + \frac{V_{DS}}{\varepsilon_c L}} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{V_{DS}^2}{2} \right] \quad -(4)$$

After integration

Velocity saturation occurs when $v = v_{sat}$.

$$I_D = W C_{ox} (V_{GS} - V_{TH} - V_{DSAT}) v_{sat} \quad -(5)$$

Equating (4) & (5).

$$V_{DSAT} = \frac{(V_{GS} - V_{TH})\,\mathcal{E}_c L}{(V_{GS} - V_{TH}) + \mathcal{E}_c L}$$

V_{DSAT} — Drain voltage at which velocity saturates.

$(V_{GS} - V_{TH})$ is small compared to $\mathcal{E}_c L$

For old xtors

$(V_{GS} - V_{TH}) << \mathcal{E}_c L$.

$$V_{DSAT} = V_{GS} - V_{TH}$$

$$I_D = \frac{\mu C_{ox}}{2} \frac{W}{L} (V_{GS} - V_{TH})^2$$

For new xtors

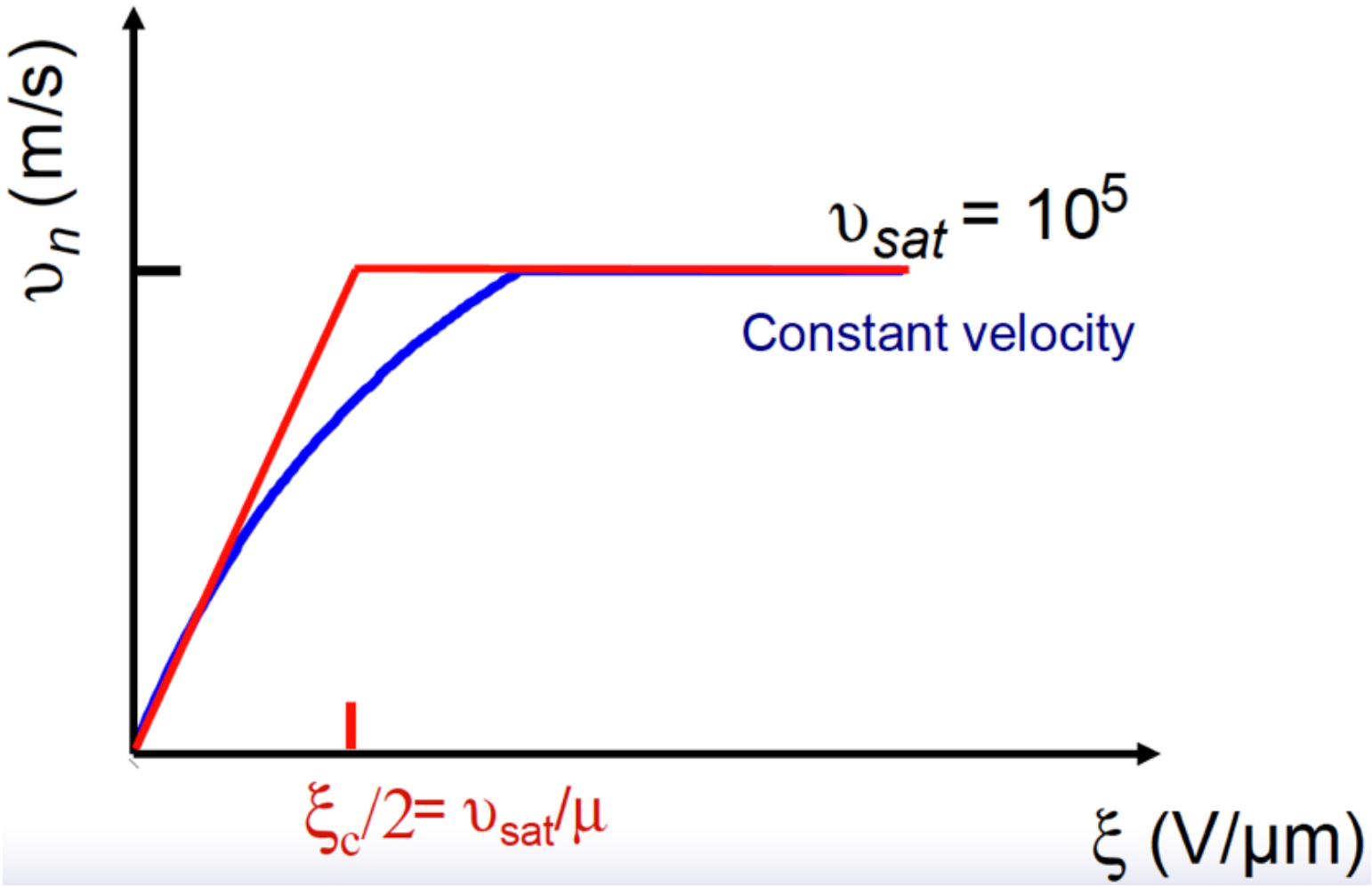
$$\varepsilon_c L << (V_{GS} - V_{TH})$$

$$V_{DSAT} = \varepsilon_c \cdot L$$

Double check

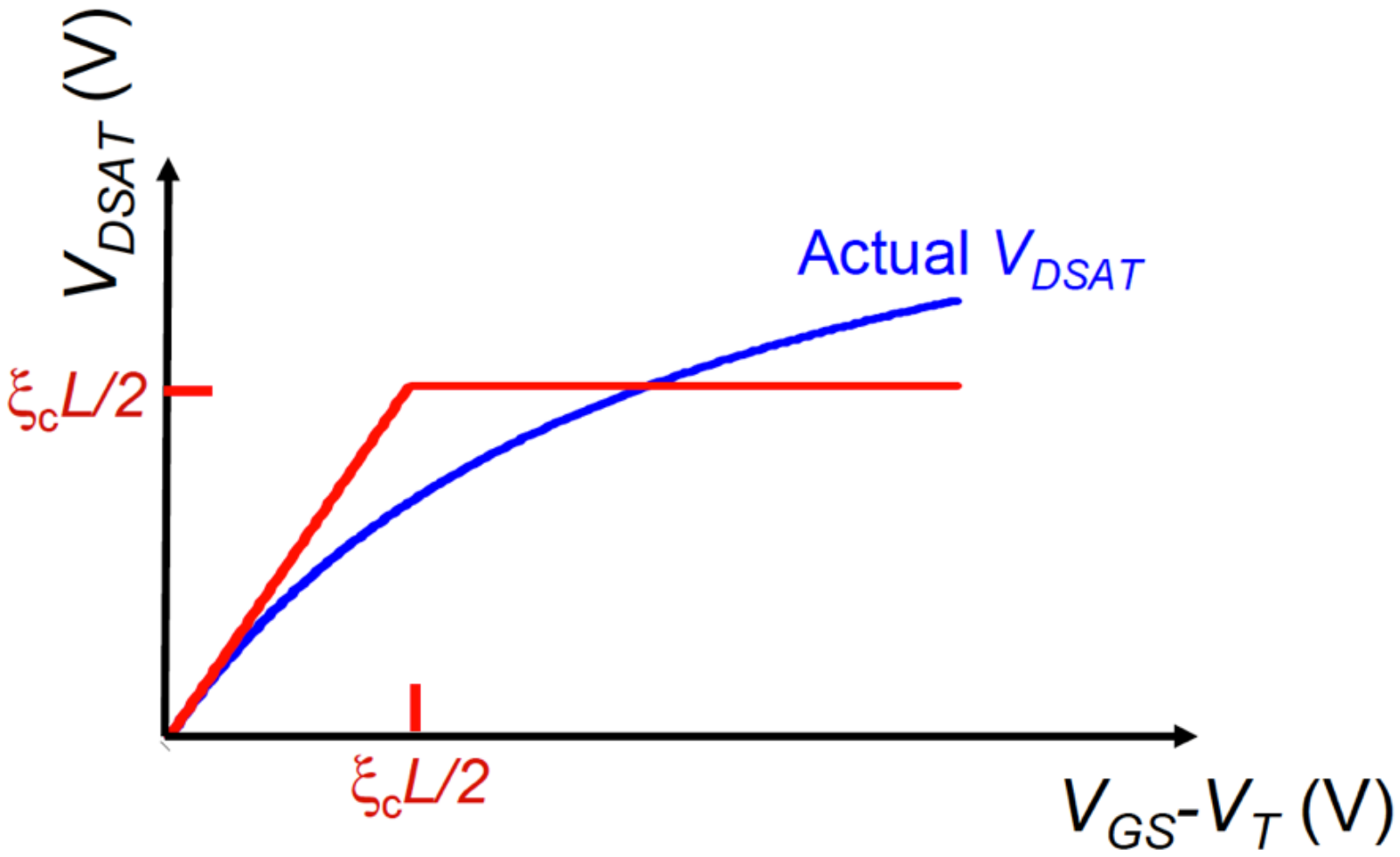
$$I_D = \frac{\mu C_{ox}}{2} \frac{W}{L} \varepsilon_c (V_{GS} - V_{TH})$$

Simplified velocity saturation



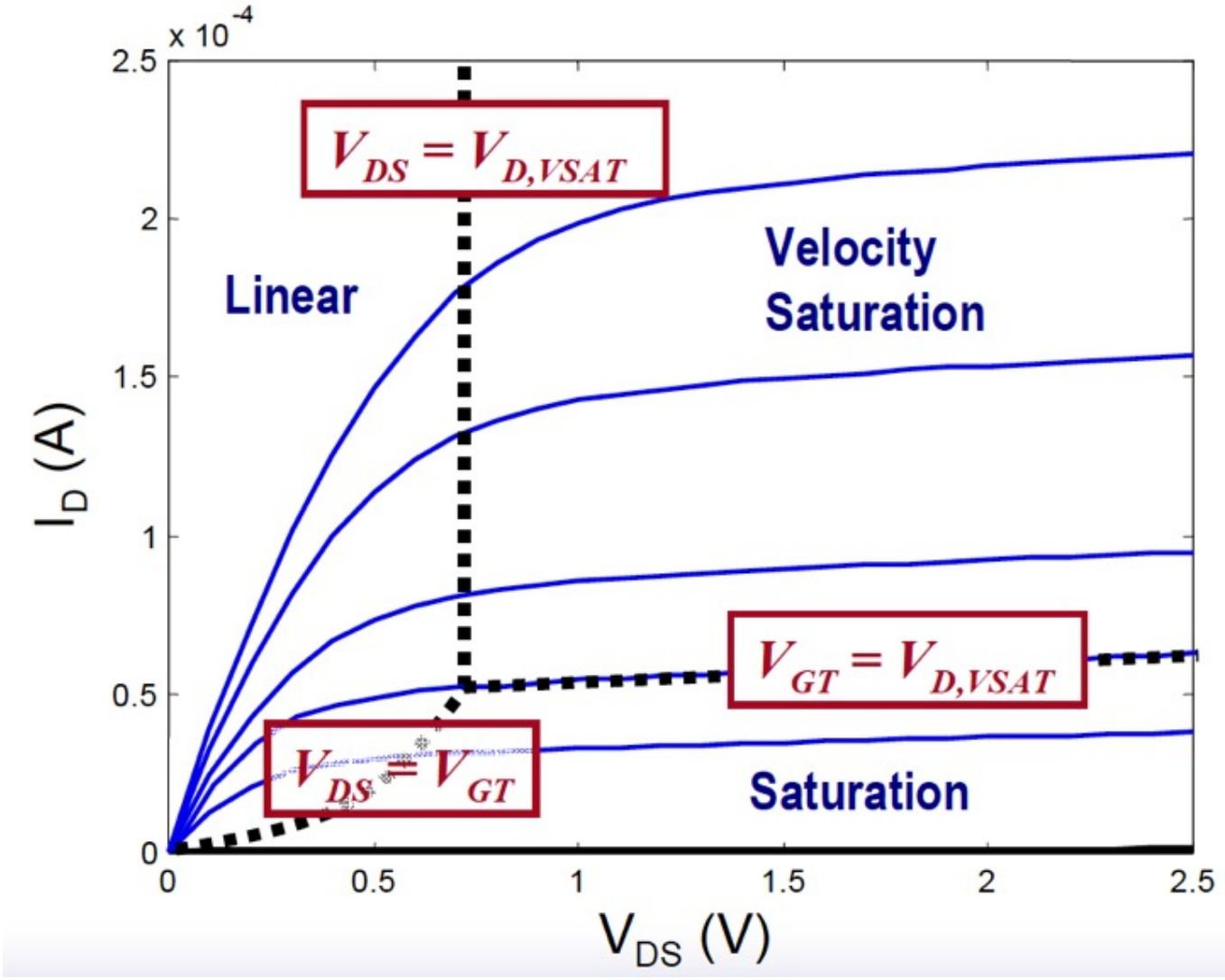
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Simplified velocity saturation



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Simplified model



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Transistor models

1. Velocity saturation model → discussed earlier.

2. alpha-power law

$$I_D \propto (V_{GS} - V_{TH})^{\alpha}$$

- α = Between 1 & 2
- Determined through measurements & simulations

③ V_T^* model → Always assume we are velocity saturated.

$$I_D = K' \frac{W}{L} \left[V_{GS} - V_{TH} - \frac{V_{DSAT}}{2} \right] V_{DSAT}$$

$\frac{V_{DSAT}}{2}$ ↑ $\mathcal{E}_c L$ at velocity saturation

$$V_T^* = V_{TH} + \frac{V_{DSAT}}{2}$$

$$I_D = K' \frac{W}{L} \left[V_{GS} - V_T^* \right] V_{DSAT}$$

④ Unified Model for manual analysis.

$$V_{GT} = V_{GS} - V_{TH}.$$

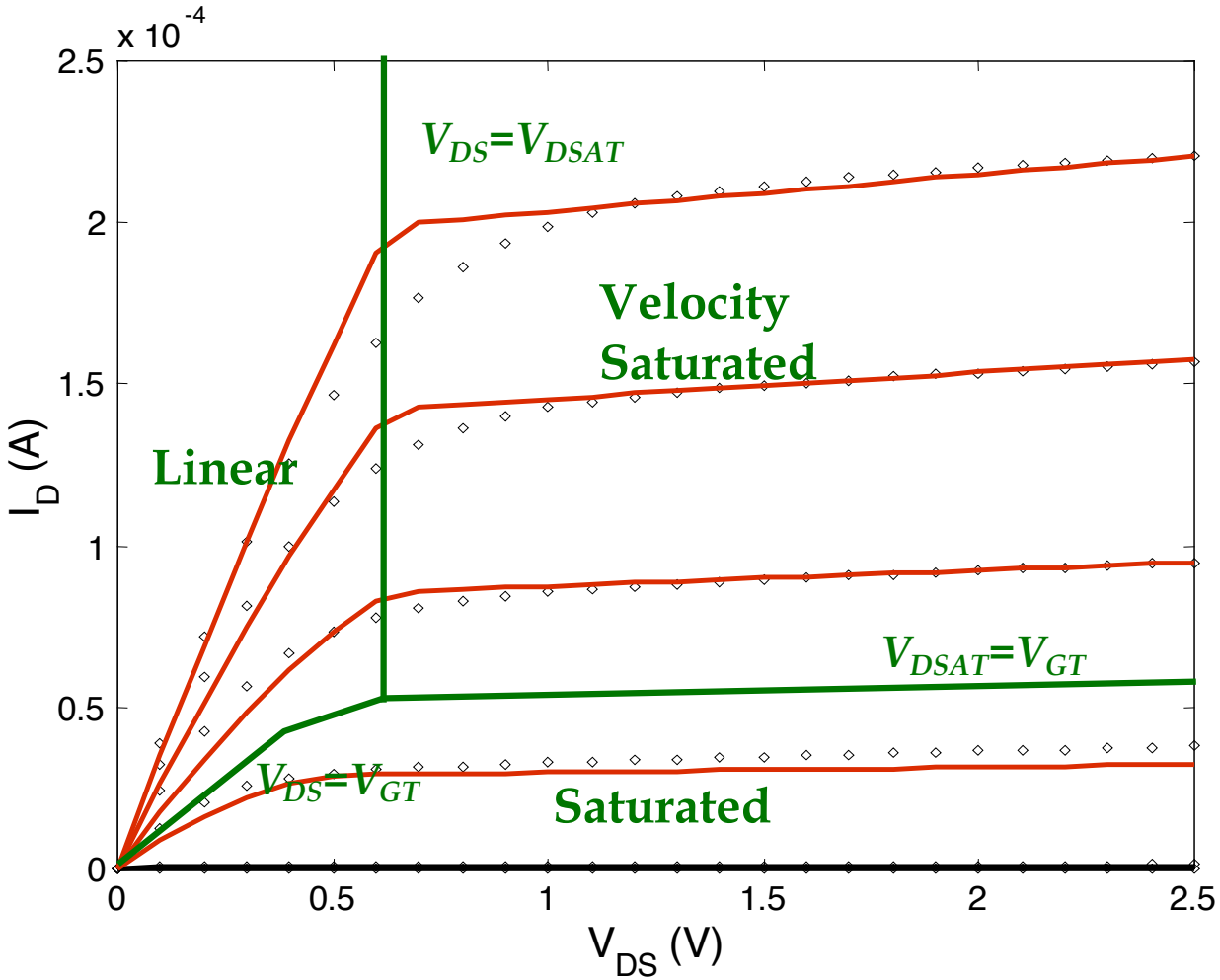
For $V_{GT} < 0$ $I_D = 0$

For $V_{GT} \geq 0$

$$I_D = K_n' \frac{W}{L} \left[V_{GT} \cdot V_{DS_{eff}} - \frac{V_{DS_{eff}}^2}{2} \right] (1 + \lambda V_{DS})$$

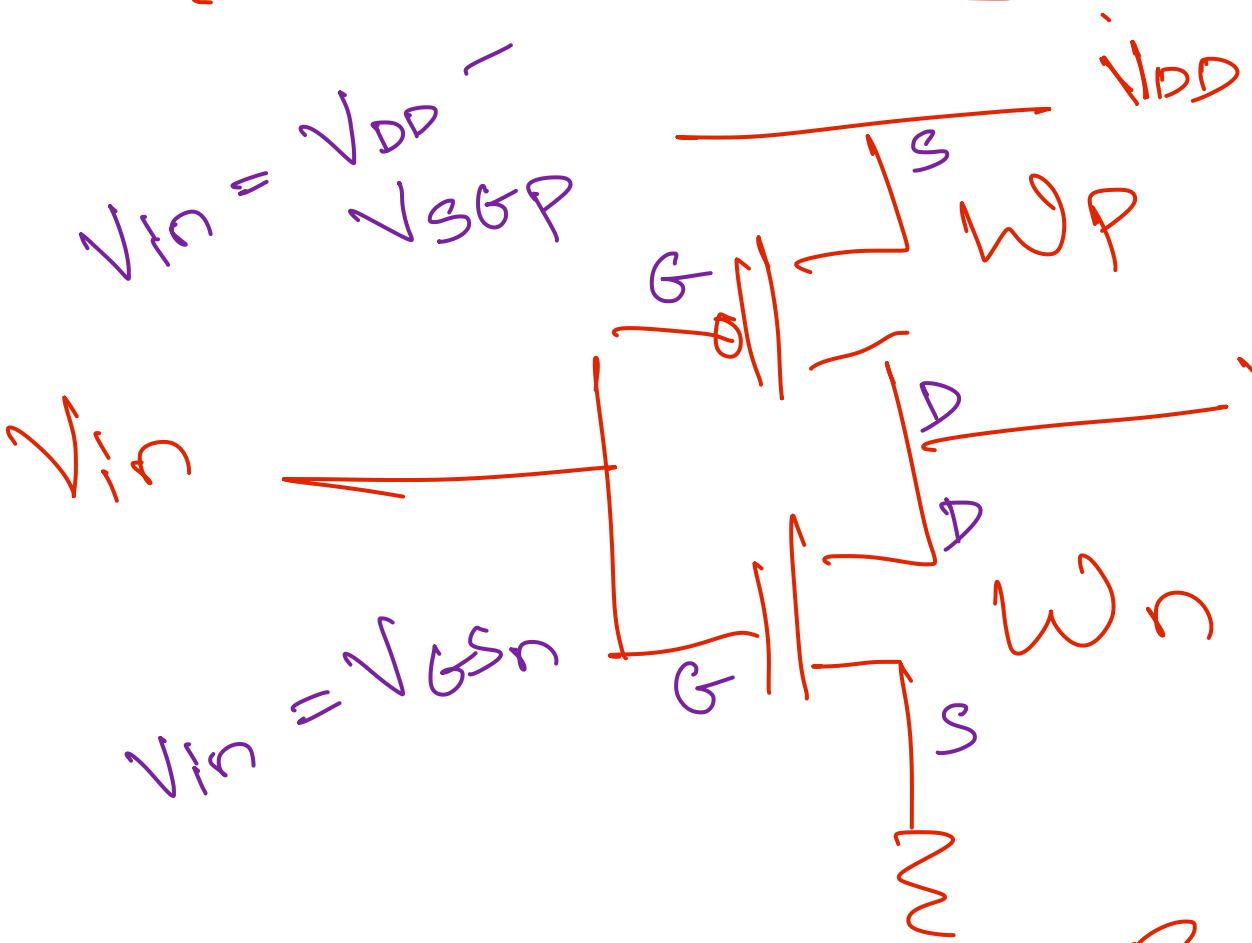
with $V_{DS_{eff}} = \min(V_{GT}, V_{DS}, V_{D_{sat}})$

Unified Model versus SPICE



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CMOS inverter

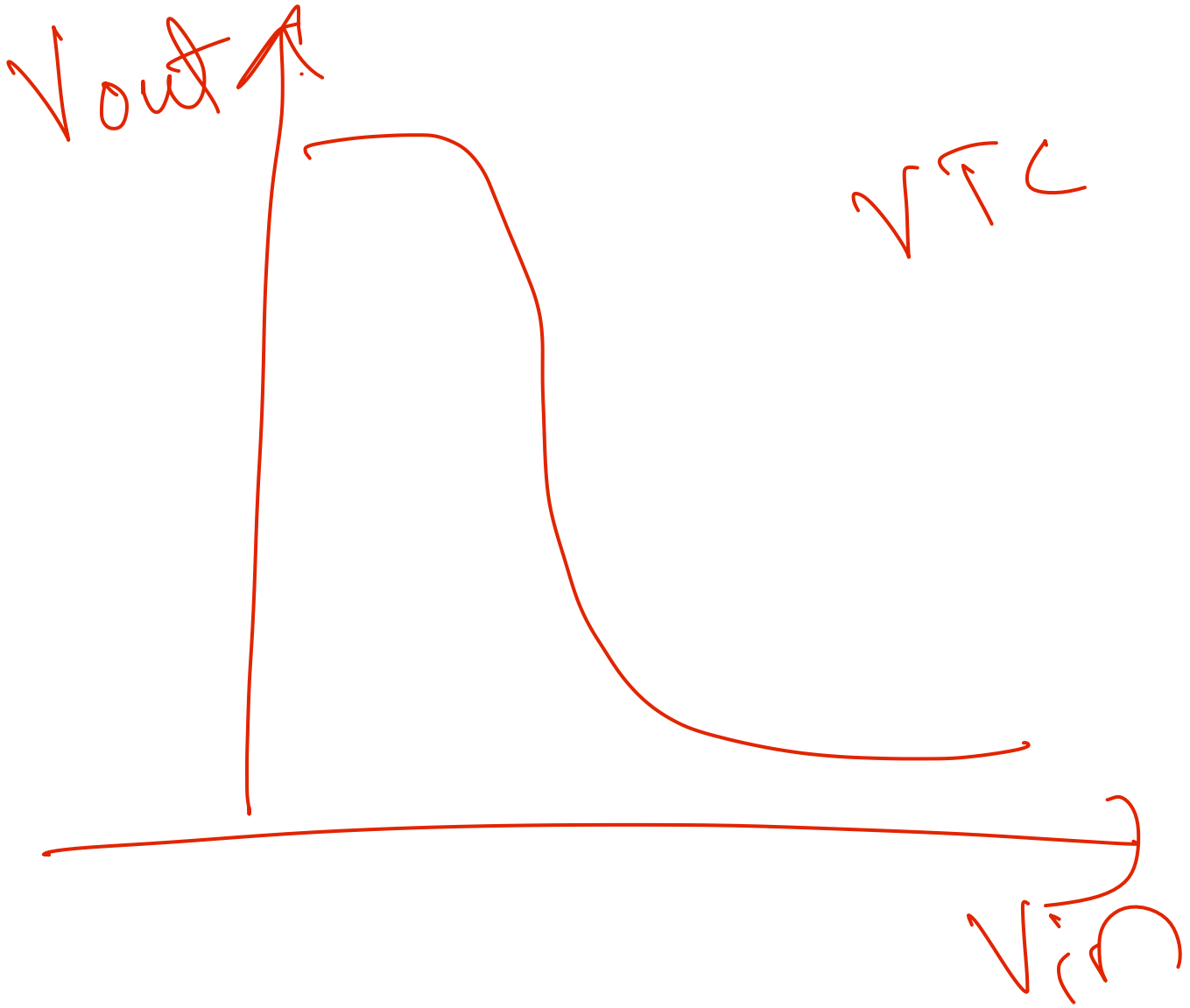


$V_{out} = V_{DSn}$

$V_{out} = V_{DD} - V_{SDP}$

$W_p = \beta \cdot W_n$

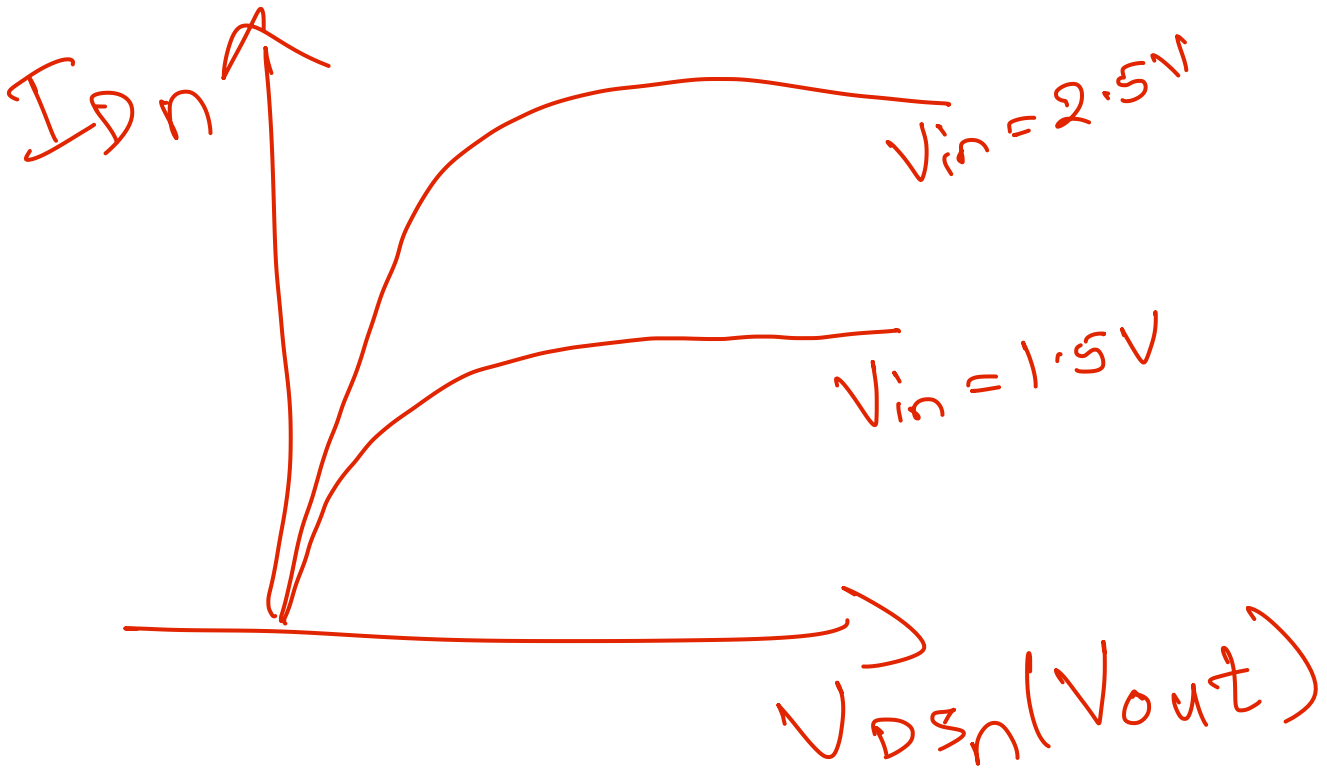
β varies between 1 & 3. We had assumed 2.


Vout
VTC
Vin

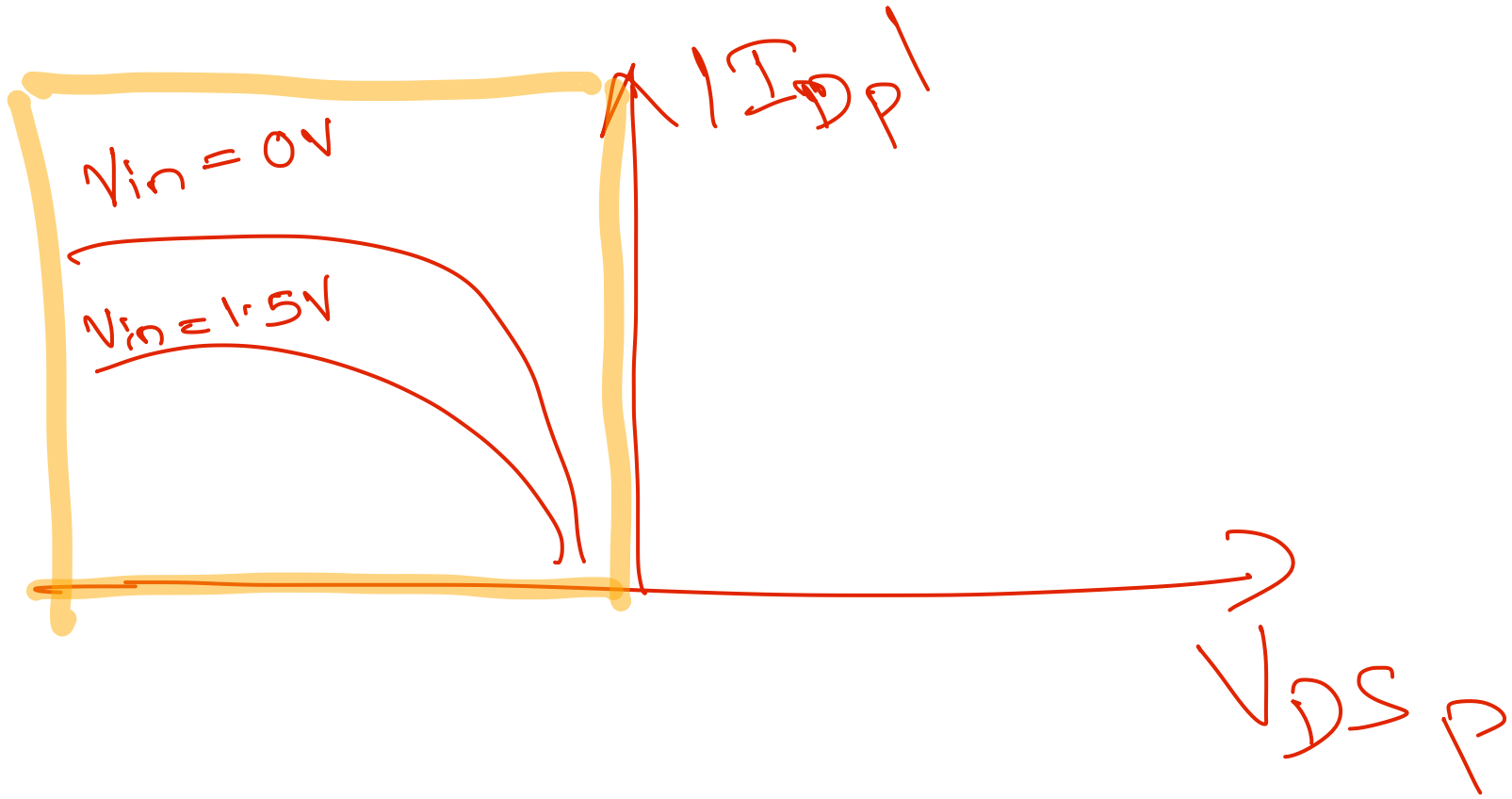
DC operation of inverter.

$$I_{Dn} = I_{Dp}$$

NMOS & PMOS IV characteristics.



NMOS x for.


$|I_{Dp}|$
$V_{in} = 0V$
$V_{in} = 1.5V$
$V_{DS\,p}$